

A

GSE208536 GeoMx cohort

Adjacent normal
(n = 16)

ADM
(n = 16)

PDAC
(n = 16)

48 spatial regions
verified by matching GeoMx segment identifiers
with GEO sample metadata

B

Spatial validation

628
malignant-switch genes
from GSE143754

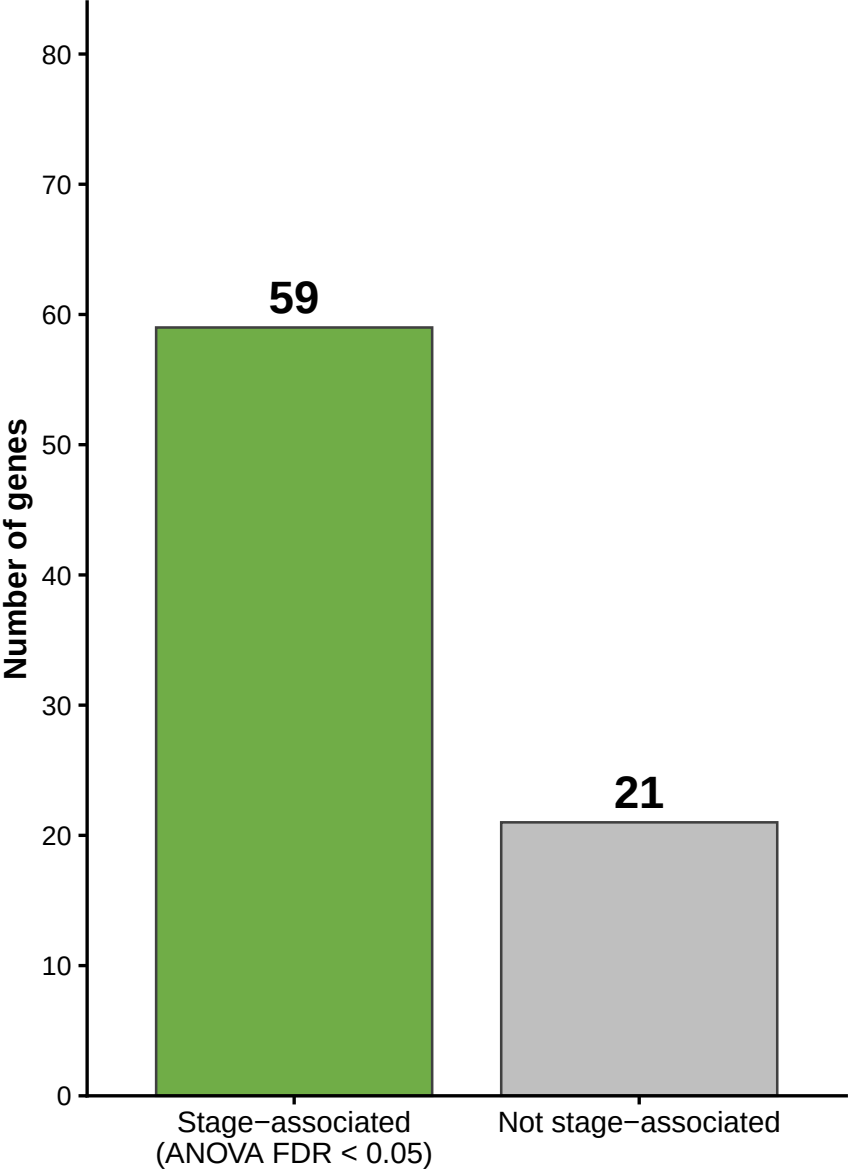
GeoMx expression
platform coverage

80 spatially validated
genes

Genes were retained only when represented
on the GeoMx expression platform.

C

ANOVA stage-association results



D

Ordinal association classification

